

Observer Representation

One Pullback, Two Pushforwards. The Geometry a Consumer Induces, Its Rate–Content Region, and Its Decision Cost

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September 2026. *Draft 0.1*. Every statement below is labeled proved, defined, or protocol. Nothing labeled protocol is a claim.

Abstract

Two earlier papers report results that look alike but stand on different ground. One characterizes the rate and conditional content of a stored description whose distortion is scored on a projection of the source. The other predicts economic choice frequencies from a hand-coded map of scenarios into nine attributes and a Mahalanobis cost. A skeptical reader can say these are two good but separate models. This paper gives the construction under which they are two applications of one object. An observer is a consumer map, an output metric, and a budget. The consumer and the output metric pull the output geometry back to the source and induce, at every regular point, a positive semidefinite read operator whose kernel is the set of directions the observer cannot distinguish. We prove that the read operator, the quotient of the source by observational indistinguishability, a global consumer-relative distortion, a nonnegative transfer regret when a representation built for one consumer is reused by another, and a consumer-relative rate–content region all follow from the triple under stated regularity. The rate–content region of the coding paper is the region obtained when the consumer is the projection of an augmented Gaussian source onto its first component. The decision cost of the economics paper is the pullback distance obtained when the consumer is the scenario encoding and the output metric is the inverse covariance. We state exactly what the high-stakes ultimatum data establish about that specialization, which is a rejection of stake-scaling invariance and not a threshold. We then give a non-reducibility theorem for the budgeted observer, whose menu-selected read subspace violates regularity and so lies outside every random-utility model, a protocol for recovering the decision geometry from individual-level multi-task data with the sample bound the recovery cliff imposes, and a behavioral game equilibrium defined without circularity, with existence, a correspondence with Nash equilibrium on a stated utility class, hardness inherited from Nash, and one closed-form case in which giving weight to a previously unread direction moves the best response.

1 Introduction

A theory earns the name when its parts are consequences of one construction rather than a family resemblance. The observation-theory program has produced two results that a careful reader can hold apart. The first is a rate–content region for a description of a Gaussian source when the consumer of the description reads only one component and a third party retains context. The second is a nine-dimensional geometric model of economic choice, calibrated on bargaining games and evaluated on lottery problems, whose monetary dimension turned out to be unread at low stakes

and whose stake-scaling corollary was then rejected on high-stakes data. Both papers say the word consumer. That is not yet a shared theory.

This paper supplies the construction and proves what it gives. The starting point is the observer triple, a consumer map from the source to an output space, a metric on the output space, and a budget. The only thing the theory ever computes with is the pullback of the output metric through the consumer. From that pullback we obtain the equivalence classes of states the observer cannot distinguish, a degenerate geometry on the source whose null directions are those classes, a global notion of consumer-relative distortion, and a nonnegative regret that a representation built for one consumer pays when a second consumer reads it. A consumer-relative rate–distortion function and a rate–content region are then definitions on top of that distortion, and we prove the properties they inherit.

The two earlier papers are then specializations. The coding paper’s region is the rate–content region when the source is a Gaussian pair, the consumer is the projection onto the first component, and the retained context is a noisy copy of the second. The economics paper’s cost is the pullback distance when the consumer is the scenario encoding and the output metric is the inverse covariance, and its inactive monetary dimension is a direction in the kernel of the pullback. The reader who wanted the two to be corollaries of one theorem gets exactly that, with the conditions written down.

We are careful about what is not proved. Conditional mutual information is not the average of a read operator. The link between information and geometry is local and passes through the Fisher metric, and we state it as a local theorem with hypotheses. The high-stakes data reject stake-scaling invariance. They do not estimate a threshold at which the monetary direction enters a budgeted subspace, and we do not claim one. The equilibrium concept we define uses an exogenous target, because a target that moves with the candidate equilibrium makes every profile an equilibrium. The complexity statements are counts of evaluations for the search that was run, not asymptotic claims about a search that was not.

2 The Observer Triple and What It Induces

2.1 Setting

Assumption 1 (Regularity). $\mathcal{M}$ is a C^1 manifold of dimension d , or an open subset of $\mathbb{R}^d$, carrying a Borel probability measure μ . $\mathcal{Y}$ is a C^1 manifold of dimension m with a continuous Riemannian metric G , so that $G(y)$ is a symmetric positive definite form on $T_y\mathcal{Y}$ for every y . The consumer $C : \mathcal{M} \rightarrow \mathcal{Y}$ is C^1 with Jacobian $J_C(x) : T_x\mathcal{M} \rightarrow T_{C(x)}\mathcal{Y}$. A point x is regular if $\text{rank } J_C$ is constant on a neighborhood of x . The budget B is either a rank cap $k \in \{1, \dots, d\}$ on representations or a rate cap $R \geq 0$ in bits per symbol.

Definition 1 (Observer). An observer is a triple $O = (C, G, B)$ satisfying Assumption 1.

2.2 The representation theorem

Theorem 1 (Observer representation). *Let $O = (C, G, B)$ be an observer. Then the following hold.*

- (a) *Indistinguishability. The relation $x \sim_C x'$ if and only if $C(x) = C(x')$ is an equivalence relation on $\mathcal{M}$ whose classes are the fibers $C^{-1}(y)$. At a regular point x with $\text{rank } J_C(x) = r$, the fiber through x is, near x , an embedded submanifold of dimension $d - r$ with tangent space $\ker J_C(x)$. If C is a submersion, the quotient $\mathcal{M} / \sim_C$ with the quotient topology is homeomorphic to the open set $C(\mathcal{M}) \subseteq \mathcal{Y}$, hence is a manifold of dimension m .*

(b) Pullback geometry. *The read operator*

$$P_C(x) = J_C(x)^\top G(C(x)) J_C(x) \quad (1)$$

is symmetric and positive semidefinite on $T_x\mathcal{M}$ for every x , with $\ker P_C(x) = \ker J_C(x)$ and $\text{rank } P_C(x) = \text{rank } J_C(x)$. It is positive definite at x if and only if $J_C(x)$ is injective. It is the pullback C^*G , a possibly degenerate Riemannian metric on $\mathcal{M}$, and at regular points its null directions are the fiber tangents of part (a). If P_C is μ -integrable, the averaged operator $\bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(X)]$ is positive semidefinite and $v \in \ker \bar{P}_{C,\mu}$ if and only if $v \in \ker P_C(x)$ for μ -almost every x .

(c) Consumer-relative distortion. Let d_G be the geodesic distance of $(\mathcal{Y}, G)$ and define the global distortion $D_C(x, \hat{x}) = d_G(C(x), C(\hat{x}))^2$. Then $D_C \geq 0$, $D_C(x, \hat{x}) = 0$ if and only if $x \sim_C \hat{x}$, and D_C is a function on $(\mathcal{M}/\sim_C)^2$. If moreover C is C^2 and G is C^1 , then for every x and every tangent δ ,

$$D_C(x, x + \delta) = \delta^\top P_C(x) \delta + o(\|\delta\|^2). \quad (2)$$

(d) Transfer regret. Let $\mathcal{Q}_B$ be a class of representations $Q = (q_{W|X}, \hat{x}(\cdot))$ admissible at budget B , and write $D_C(Q) = \mathbb{E}[D_C(X, \hat{x}(W))]$. Let $Q_1^* \in \arg \min_{\mathcal{Q}_B} D_{C_1}$ exist. Then

$$L_B(C_1 \rightarrow C_2) = D_{C_2}(Q_1^*) - \inf_{Q \in \mathcal{Q}_B} D_{C_2}(Q) \geq 0, \quad (3)$$

with equality if and only if Q_1^* is also C_2 -optimal in $\mathcal{Q}_B$.

Proof. (a) Reflexivity, symmetry, and transitivity are those of equality in $\mathcal{Y}$, and the class of x is $C^{-1}(C(x))$ by definition. At a regular point the constant-rank theorem [1] gives charts in which C is the linear map $(u_1, \dots, u_d) \mapsto (u_1, \dots, u_r, 0, \dots, 0)$, so the fiber is locally the coordinate slice $u_1 = \dots = u_r = \text{const}$, an embedded submanifold of dimension $d - r$ whose tangent space is the kernel of the linearization, which is $\ker J_C(x)$. If C is a submersion it is an open map, so $C(\mathcal{M})$ is open in $\mathcal{Y}$. The induced map $\bar{C} : \mathcal{M}/\sim_C \rightarrow C(\mathcal{M})$ is a bijection by construction, continuous because C is, and open because a set U is open in the quotient exactly when its preimage $\pi^{-1}(U)$ is open in $\mathcal{M}$, in which case $\bar{C}(U) = C(\pi^{-1}(U))$ is open. A continuous open bijection is a homeomorphism.

(b) For tangents u, v , $u^\top P_C(x)v = \langle J_C u, J_C v \rangle_{G(C(x))}$, which is symmetric and satisfies $v^\top P_C v = \|J_C v\|_G^2 \geq 0$. Since $G \succ 0$, $v^\top P_C v = 0$ if and only if $J_C v = 0$, which gives the kernel, the rank, and the definiteness statement. The formula (1) is the coordinate expression of the pullback C^*G . At regular points the kernel is the fiber tangent by (a). For the average, $v^\top \bar{P} v = \mathbb{E}_\mu[v^\top P_C(X)v]$ is the integral of a nonnegative function, so it vanishes if and only if the integrand vanishes μ -almost everywhere.

(c) d_G is a metric on $\mathcal{Y}$, so $D_C \geq 0$ with equality if and only if $C(x) = C(\hat{x})$, and D_C depends on its arguments only through their classes. For the expansion, let $\gamma(t) = C(x + t\delta)$, a C^2 curve in $\mathcal{Y}$ with $\gamma'(0) = J_C(x)\delta$. On a normal neighborhood of $\gamma(0)$ the squared geodesic distance $d_G(\gamma(0), \gamma(t))^2$ is C^2 in t with $d_G(\gamma(0), \gamma(t))^2 = \|\gamma'(0)\|_G^2 t^2 + o(t^2)$, which is the standard second-order expansion of the squared distance along a C^2 curve. Substituting $\gamma'(0)$ gives (2).

(d) The infimum on the right of (3) is over a set that contains Q_1^* , so it is at most $D_{C_2}(Q_1^*)$. Equality holds exactly when Q_1^* attains the infimum. $\square$

Remark. Part (b) does not say that P_C is positive definite, and it is not in the cases of interest. Low rank is the content of the theory. Part (c) is a global definition with a local quadratic expansion, and the expansion is what every trace formula in the earlier papers used. Part (d) replaces a difference of operators, which has no sign, by a regret, which does.

2.3 The linear quadratic case of transfer regret

The regret (3) has a closed form when consumers are linear, the local expansion is exact, and representations are projections.

Proposition 1 (Transfer regret for projections). *Let X be a random vector in $\mathbb{R}^d$ with $\mathbb{E}[XX^\top] = \sigma^2 I$, let C_1, C_2 be linear with read operators P_1, P_2 constant in x , and let $\mathcal{Q}_B$ be the class of representations that keep an orthogonal projection Π_U onto a B -dimensional subspace U and reconstruct $\hat{x} = \Pi_U x$. Then $D_{C_i}(U) = \sigma^2 (\text{tr } P_i - \text{tr}(\Pi_U P_i))$, the C_i -optimal subspace is a top- B eigenspace U_i of P_i , and*

$$L_B(C_1 \rightarrow C_2) = \sigma^2 \left(\sum_{j=1}^B \lambda_j(P_2) - \text{tr}(\Pi_{U_1} P_2) \right) \geq 0, \quad (4)$$

where $\lambda_1 \geq \dots \geq \lambda_d$ are the eigenvalues of P_2 . Equality holds if and only if U_1 is a top- B eigenspace of P_2 . If $P_2 = \sum_j \lambda_j u_j u_j^\top$, then $\text{tr}(\Pi_{U_1} P_2) = \sum_j \lambda_j \|\Pi_{U_1} u_j\|^2$, so the regret is a weighted sum of the misalignment of the read directions of C_2 with the subspace kept for C_1 .

Proof. With $\delta = (I - \Pi_U)x$ and a constant read operator, $D_{C_i}(U) = \mathbb{E}[\delta^\top P_i \delta] = \text{tr}(P_i(I - \Pi_U)\sigma^2 I(I - \Pi_U)) = \sigma^2 \text{tr}((I - \Pi_U)P_i)$, using $(I - \Pi_U)^2 = I - \Pi_U$ and cyclicity. Minimizing over U is maximizing $\text{tr}(\Pi_U P_i)$ over rank- B orthogonal projections, and Ky Fan's maximum principle [2] gives the value $\sum_{j \leq B} \lambda_j(P_i)$, attained exactly on top- B eigenspaces. Substituting into (3) gives (4), and the sign and the equality case are Ky Fan's again. The last identity is $\text{tr}(\Pi_{U_1} u_j u_j^\top) = u_j^\top \Pi_{U_1} u_j = \|\Pi_{U_1} u_j\|^2$. $\square$

Remark. When $\mathbb{E}[XX^\top] = \Sigma_X$ is not isotropic, whiten first. With $\tilde{x} = \Sigma_X^{-1/2}x$ the read operators become $\tilde{P}_i = \Sigma_X^{1/2} P_i \Sigma_X^{1/2}$ and the proposition applies to them with $\sigma^2 = 1$. The subspaces are then optimal in the whitened coordinates, which is what a code built on the source covariance keeps.

2.4 Consumer-relative rate-distortion and the rate-content region

Definition 2 (Consumer-relative rate-distortion function). For a source $X \sim \mu$ on $\mathcal{M}$ and an observer O , the consumer-relative rate-distortion function is

$$R_C(D) = \inf \{ I(X; W) : q_{W|X}, \hat{x}(\cdot), \mathbb{E}[D_C(X, \hat{x}(W))] \leq D \}. \quad (5)$$

Proposition 2. R_C is nonincreasing and convex in D on the set where it is finite. If P_C is constant and equal to P , and the expansion (2) is exact, then R_C is the rate-distortion function of X under the weighted quadratic distortion $(x - \hat{x})^\top P(x - \hat{x})$, and for Gaussian X it is obtained by reverse water-filling on the eigenvalues of $P^{1/2} \Sigma_X P^{1/2}$.

Proof. Monotonicity is immediate from enlarging the constraint set. Convexity is the time-sharing argument for a general distortion measure [3]. If $(q_1, \hat{x}_1)$ and $(q_2, \hat{x}_2)$ meet distortions D_1, D_2 , the mixture channel that uses the first with probability λ and the second otherwise, with the choice encoded in W , meets $\lambda D_1 + (1 - \lambda) D_2$, and mutual information is convex in the conditional law for a fixed source, so the mixture's rate is at most $\lambda I_1 + (1 - \lambda) I_2$. The quadratic specialization is the definition, and the Gaussian statement is the change of variables $\tilde{x} = P^{1/2}x$ that turns the weighted quadratic distortion into squared error, after which the classical reverse water-filling [3] applies on the covariance of $\tilde{x}$. $\square$

Definition 3 (Rate–content region). Let (X, S) be jointly distributed, with S the context retained by a third party. The consumer–relative rate–content region at distortion D is

$$\mathcal{R}_C(D) = \left\{ (R, L) : \exists (q_{W|X}, \hat{x}) \text{ with } R \geq I(X; \hat{X}), L \geq I(X; \hat{X} | S), \mathbb{E}[D_C(X, \hat{X})] \leq D \right\}, \quad (6)$$

where $\hat{X} = \hat{x}(W)$ and the joint law of (X, S, W) has W conditionally independent of S given X .

The region is a definition, and two facts give it operational content. The first is a lemma that lets the reproduction live in the output space, which is where a coding theorem places it.

Lemma 1 (Reproduction in the output space). *Let $C : \mathcal{M} \rightarrow \mathcal{Y}$ be surjective. For every test channel $q_{W|X}$ and every reproduction map $\hat{y} : \mathcal{W} \rightarrow \mathcal{Y}$ there is a reproduction map $\hat{x} : \mathcal{W} \rightarrow \mathcal{M}$ with $C \circ \hat{x} = \hat{y}$, and conversely every $\hat{x}$ induces $\hat{y} = C \circ \hat{x}$, with the same distortion $\mathbb{E}[d_G(C(X), \hat{y}(W))^2]$ and the same values of $I(X; W)$ and $I(X; W | S)$. The region (6) is therefore unchanged if the reproduction is taken in $\mathcal{Y}$ and the distortion is $d_G(C(X), \hat{Y})^2$.*

Proof. Choose $\hat{x}(w)$ in the fiber $C^{-1}(\hat{y}(w))$, which is nonempty by surjectivity. The information quantities depend on (X, S, W) only, and the distortion depends on the reproduction only through $C \circ \hat{x}$. $\square$

The second is the coding theorem of the rate–content paper, which we restate in the vocabulary of (6) and verify against its operational definitions in Section 3. We now record the one rigorous link between the information quantities in (6) and the read operator. It is local.

Theorem 2 (Local Fisher bridge). *Let the channel from X to W read x only through the consumer, so that $q_{W|X=x} = \kappa(\cdot | C(x))$ for a family $\{\kappa(\cdot | y)\}_{y \in \mathcal{Y}}$ of densities that is C^2 in y with Fisher information $F(y) = \mathbb{E}_{\kappa(\cdot | y)}[\nabla_y \log \kappa \nabla_y \log \kappa^\top]$ finite and continuous, under the usual conditions permitting differentiation under the integral. Then for every x and tangent δ ,*

$$D_{\text{KL}}(q_{W|X=x+\delta} \| q_{W|X=x}) = \frac{1}{2} \delta^\top J_C(x)^\top F(C(x)) J_C(x) \delta + o(\|\delta\|^2), \quad (7)$$

that is, the local divergence geometry of the channel on $\mathcal{M}$ is the read operator of the observer $(C, F, \cdot)$ whose output metric is the Fisher metric of the channel.

Proof. For a C^2 family the Kullback–Leibler divergence between $\kappa(\cdot | y + \eta)$ and $\kappa(\cdot | y)$ equals $\frac{1}{2} \eta^\top F(y) \eta + o(\|\eta\|^2)$, which is the standard second-order expansion of the divergence with vanishing first-order term [4]. Apply it with $y = C(x)$ and $\eta = C(x + \delta) - C(x) = J_C(x) \delta + o(\|\delta\|)$, and use continuity of F to absorb the remainder. $\square$

Remark. Theorem 2 is the correct form of the statement that information theory is the population version of the read operator. The quantity $I(X; W | S)$ is an expectation of divergences between conditional laws, $\mathbb{E}[D_{\text{KL}}(p_{W|X,S} \| p_{W|S})]$, and it is not an average of P_C . What the theorem gives is that the divergence between the channel outputs at two nearby source points is, to second order, the read-operator quadratic form with the Fisher metric as G . Every global information quantity in (6) is built from such divergences, and no more than that is claimed.

3 First Pushforward. The Rate–Content Region with Encoder–Observed Context

The coding paper [5] studies an encoder that observes a source and stores a description $M = f_n(X^n)$ at rate $R_n = \frac{1}{n} \log |\mathcal{M}_n|$ for a decoder g_n that sees M alone, while a third party retains a noisy

copy S^n of a context variable and never decodes. Its three operational quantities are the rate R_n , the per-letter distortion D_n , and the conditional content $L_n = \frac{1}{n}H(M | S^n)$, and a triple (R, L, D) is achievable when a code sequence meets all three in the limit superior. Its single-letter characterization is, for a discrete memoryless pair (X, S) with bounded distortion, and again for the jointly Gaussian source with quadratic distortion on Y at every $0 < D < 1$,

$$\mathcal{RW}(D) = \bigcup_{p(\hat{x}|x): \mathbb{E}d(X, \hat{X}) \leq D} \{(R, L) : R \geq I(X; \hat{X}), L \geq I(X; \hat{X} | S)\}, \quad (8)$$

with the Markov chain $\hat{X} - X - S$ imposed because the description is formed without access to S . The content endpoint $\min I(X; \hat{X} | S)$ is Steinberg's common-reconstruction functional [6] as an information quantity, and the coding paper is explicit that the operational roles differ. In Steinberg's problem S is decoder side information that must not affect reproducibility. In the coding paper S is held by a party that never serves the decoder, and no common-reconstruction constraint appears in the code definition.

Corollary 1 (The coding region is a consumer-relative region). *Take $\mathcal{M} = \mathbb{R}^d$ with μ a centered Gaussian law, the consumer the linear functional $C(x) = w^\top x = Y$ with $\mathcal{Y} = \mathbb{R}$ and $G = 1$, the context $V = \gamma^\top x / \sigma_\gamma$ a second linear functional with $\mathbb{E}[YV] = \rho$, and the retained copy $S = V + U$ with $U \sim \mathcal{N}(0, \tau^2)$ independent of X . Then $D_C(x, \hat{x}) = (w^\top x - w^\top \hat{x})^2$, the read operator is the constant rank-one form $P_C = ww^\top$ whose kernel $w^\perp$ contains the context direction unless γ is parallel to w , and the region of Definition 3 at distortion D equals the operational region (8) of the coding paper for $0 < D < 1$.*

Proof. With C linear, $J_C = w^\top$ and (1) gives $ww^\top$ exactly, with no remainder in (2), and $\ker P_C = w^\perp$. Definition 3 imposes W conditionally independent of S given X , which is the Markov chain $\hat{X} - X - S$ of the coding paper, and its two information inequalities are those of (8). Its distortion is $\mathbb{E}[(w^\top X - w^\top \hat{X})^2] \leq D$, and by Lemma 1 with the surjective $C = w^\top$ this is the coding paper's constraint $\mathbb{E}[(Y - \hat{Y})^2] \leq D$ on an output-space reproduction. The coding paper's pair-sufficiency theorem shows that the test channel may depend on x only through $T = (Y, V)$ without loss, and its Gaussian operational theorem identifies the single-letter union with the closure of the achievable triples for $0 < D < 1$. The two regions therefore coincide on that range. $\square$

The verification behind the corollary is the following list, item by item against the coding paper's problem statement. The encoder sees X^n only, which is the source law μ of Definition 3. The decoder sees M only, which is the conditional independence of W and S given X . The distortion is on Y alone, which is D_C for the projection consumer. The content is $H(M | S^n)/n$, whose single-letter form is the second inequality. The third party never reconstructs, so no common-reconstruction constraint enters, and none is needed in Definition 3. Two hypotheses of the coding theorems are not in Definition 3 and must be carried by the instance. The discrete theorem needs a bounded distortion, and the Gaussian theorem needs $0 < D < 1$ with the quadratic distortion handled by its own compactness argument.

Remark. The corollary is an identification of definitions, now verified. The theorems of the coding paper, the closed form for $L(D)$, the coupled two-water-level frontier, and non-determination under reduction of (Y, V) to (Y, S) , are theorems about this instance and are not re-derived here. What the corollary adds is the reading. The kernel of the read operator is everything orthogonal to the read direction w , the context direction γ lies in that kernel except when it is parallel to w , the content L is an information quantity about a variable the consumer never reads, and the non-determination corollary of that paper says that L is not a functional of the law of $(C(X), S)$ alone. That last point

is the strongest evidence in the program that the quotient of Theorem 1(a) is not the whole story once a third party retains context, because the encoder’s access to the unread direction V enters the answer.

4 Second Pushforward. Geometric Decision Cost

The economics paper [7] encodes each task as a reference point and a menu of alternatives in $\mathbb{R}^9$ and scores an alternative by the Mahalanobis distance of its displacement from the reference under a diagonal covariance Σ . Three variances were fitted on nine game targets, and the model matched sixteen aggregate targets within stated tolerances with a mean absolute error of 2.70 percentage points.

Corollary 2 (Decision cost is a pullback distance). *Let $\mathcal{A}$ be the set of scenario alternatives of a task, together with its reference r . Let the consumer be the scenario encoding $C : \mathcal{A} \cup \{r\} \rightarrow \mathbb{R}^9$ of the economics paper, and let the output metric be $G = \Sigma^{-1}$ on the active coordinates, extended by zero on the inactive coordinates. Then the decision cost of alternative a is the pullback distance*

$$c(a)^2 = (C(a) - C(r))^\top \Sigma^{-1} (C(a) - C(r)), \quad (9)$$

the game prediction $\arg \min_a c(a)$ and the lottery rule $P(A) \propto \exp(-c_A/T)$ are the choice rules of that paper, and the inactive dimensions are the coordinate directions in the kernel of G . When the scenario space is a continuous manifold and C is C^1 , the local metric on it is $J_C^\top \Sigma^{-1} J_C$, which is (1).

Proof. Equation (9) is the definition of the Mahalanobis cost in the economics paper with the displacement written as a difference of consumer outputs. For discrete alternatives no Jacobian is needed and (9) is the exact pullback of the quadratic form G through C . On a continuous scenario manifold the local statement is Theorem 1(b) with $G = \Sigma^{-1}$. $\square$

Two clarifications bound the specialization. The consumer is the encoding, not the identity. The encoding is the hand-coded part of the economics paper and it is where that paper’s audit tables live. The three active dimensions are a sparsity pattern of G selected by a search over active sets. They are not a demonstrated cognitive budget. A budget in the sense of the observer triple would be a rank cap imposed on the reader and tested against the reader’s behavior under larger menus, and Section 6 says how that could be measured.

4.1 What the high-stakes data establish

Proposition 3 (Stake-scaling invariance). *Let the encoding depend on the stake $\lambda > 0$ only through the monetary coordinate, $C_\lambda(a) = C_1(a) + (\lambda - 1) e_1 e_1^\top C_1(a)$ for all alternatives and the reference. If $e_1^\top G e_1 = 0$, then $c_\lambda(a) = c_1(a)$ for every a and every λ , and every prediction of the model is invariant to the stake. Conversely, if some alternative has a monetary displacement $e_1^\top (C_1(a) - C_1(r)) \neq 0$ and $c_\lambda(a)$ is the same for two distinct values of λ , then $e_1^\top G e_1 = 0$.*

Proof. With G diagonal, $c_\lambda(a)^2 = c_1(a)^2 + (\lambda^2 - 1) G_{11} (e_1^\top (C_1(a) - C_1(r)))^2$. The forward direction sets $G_{11} = 0$. The converse divides by the nonzero monetary displacement and by $\lambda^2 - 1 \neq 0$. $\square$

The economics paper reports that on the data of Andersen, Ertac, Gneezy, Hoffman, and List [8], collected in eight villages of Meghalaya in northeast India with 458 responders and stakes from 20 to 20,000 rupees, the largest exceeding the villagers’ average yearly income, offers and rejections

vary with the stake at every conventional level. By Proposition 3 this rejects the conjunction of a stake-blind encoding with an unread monetary direction. It does not estimate the stake at which the monetary direction would enter a budget-selected subspace, and it does not show that any subspace was budget-selected. The reference implementation of the economics paper [9] normalizes the monetary coordinate to the stake, so the encoding as released is stake-blind for every G , and a model that fits both stake ranges needs a monetary coordinate that carries the absolute stake together with a finite monetary variance. Saying that the monetary direction reactivates at high stakes is an interpretation of the rejection. A threshold claim needs a model with a stake-carrying coordinate fitted across stake levels and tested against held-out levels.

5 Non-Reducibility of the Budgeted Observer

A skeptic can ask what the geometry predicts that a scalar utility with logit noise cannot. For a fixed read operator the answer is nothing. The choice rule $P(a) \propto \exp(-c(a)/T)$ with a fixed cost is a Luce model [10] with utility $-c/T$, it satisfies independence of irrelevant alternatives, and it satisfies regularity. The economics paper’s model is of this kind and we do not claim otherwise.

The claim that survives concerns the budgeted observer, whose read subspace is selected from the menu it faces. Such an observer changes what it reads when the menu changes, and the resulting stochastic choice function lies outside every random-utility model.

Definition 4 (Budgeted menu-relative choice). Let x_0 be a reference point in $\mathbb{R}^d$ and let a menu A be a finite set of alternatives with encodings $x_a \in \mathbb{R}^d$. Let $M(A) = \sum_{a \in A} (x_a - x_0)(x_a - x_0)^\top$ and let $\Pi_B(A)$ be the orthogonal projector onto a top- B eigenspace of $M(A)$. The budgeted observer chooses

$$p(a \mid A) = \frac{\exp(-\|\Pi_B(A)(x_a - x_0)\|^2)}{\sum_{b \in A} \exp(-\|\Pi_B(A)(x_b - x_0)\|^2)}. \quad (10)$$

Theorem 3 (Non-reducibility). *There exist $d = 2$, $B = 1$, a reference $x_0 = 0$, and alternatives a, b, c such that the budgeted observer of Definition 4 satisfies $p(a \mid \{a, b, c\}) > p(a \mid \{a, b\})$. Consequently (10) violates regularity, is not the choice function of any random-utility model, and in particular is not a Luce model $p(a \mid A) \propto \exp(u(x_a))$ for any utility u and any temperature.*

Proof. Take $x_a = (2, 0)$, $x_b = (0, 1)$, $x_c = (0, 3)$. For $A = \{a, b\}$, $M(A) = \text{diag}(4, 1)$, the top eigenvector is e_1 , the projected costs are 4 and 0, and $p(a \mid A) = e^{-4}/(e^{-4} + 1) = 0.0180$. For $A' = \{a, b, c\}$, $M(A') = \text{diag}(4, 10)$, the top eigenvector is e_2 , the projected costs are 0, 1, and 9, and $p(a \mid A') = 1/(1 + e^{-1} + e^{-9}) = 0.7310$. Adding the alternative c raised the probability of a from 0.018 to 0.731. Regularity, which requires $p(a \mid A') \leq p(a \mid A)$ whenever $A \subseteq A'$, is violated. Every random-utility model satisfies regularity [11, 12], because the event that a has the largest utility in A' is contained in the event that it has the largest utility in A . Luce models are random-utility models with independent Gumbel noise, so the last claim follows. $\square$

Remark. The theorem is a statement about the observer whose budget selects its read subspace from the menu. It is not a statement about the economics paper’s fitted model, which has a fixed read operator and is a Luce model. The empirical question the theorem poses is whether human readers of choice menus behave like the budgeted observer, that is, whether a decoy that changes the second moment of the menu can raise the share of an existing option. Attraction and decoy effects in the marketing and psychology literatures [13] are regularity violations of exactly this form, and the budgeted observer offers one mechanism for them. Testing that mechanism is a designed experiment, not a reanalysis of the sixteen aggregate targets. The stake rejection of Section 4 is

a different statement. It says that no stake-blind encoding fits both stake ranges. It says nothing about scalarization.

6 Protocol. Learning the Decision Geometry Instead of Coding It

Everything in this section is protocol, not result. The economics paper’s encoding assigns coordinates by hand. The observation-theory program has an instrument, the blind probe [14, 15], that recovers a consumer’s read operator from calls to the consumer alone, and a theorem, the budget cliff [15], that says recovery at an operating point is a cliff at the full dimension of the space rather than a slope. The analogue for a human decision consumer is a designed menu.

Object to be recovered. For subject i , a symmetric positive semidefinite 9×9 precision Ω_i such that $p_i(a \mid A) \propto \exp(-(x_a - x_0)^\top \Omega_i (x_a - x_0)/T)$, with the population read operator $\bar{\Omega} = \mathbb{E}_i[\Omega_i]$. The hypothesis under test is that the top eigenspace of $\bar{\Omega}$ aligns with the hand-coded active set $\{d_6, d_7, d_9\}$.

Design bound from the cliff. A symmetric 9×9 form has 45 free entries and a diagonal one has 9. A binary choice between alternatives with displacements δ_a, δ_b identifies the sign of $\delta_a^\top \Omega \delta_a - \delta_b^\top \Omega \delta_b$ and, with a temperature, its magnitude up to noise. The cliff theorem says that a probe confined to fewer than d independent directions cannot identify a hidden leading eigenspace. The per-subject menu battery must therefore span at least nine linearly independent displacement directions for a diagonal target and at least forty-five independent displacement pairs for a full one, before any allowance for noise. The sixteen aggregate targets of the economics paper span at most five directions in their game encodings, which is why that paper could not have identified the geometry even in principle.

Hierarchical model. $\Omega_i \sim \text{Wishart}(\bar{\Omega}/\nu, \nu)$, $p_i(a \mid A)$ as above, with T fixed at the economics paper’s value or estimated once for the population. The estimator of $\bar{\Omega}$ is the posterior mean, and its leading k -dimensional eigenspace $\hat{U}_k$ is compared with the hand-coded subspace $U_0 = \text{span}\{e_6, e_7, e_9\}$.

Statistic and floor. Overlap $\text{tr}(\Pi_{\hat{U}_k} \Pi_{U_0})/k$, reported as resolution (overlap—chance)/(1—chance) with chance $k/9$ for a random k -dimensional subspace, together with the rank certificate of the readscape instrument [14] on $\hat{\bar{\Omega}}$. In the scalar Euclidean case the population read operator is the active-subspace matrix [16], and the same estimators apply.

Bars. To be frozen before any subject is run. A resolution bar, a rank bar, and an anti-vacuity bar requiring that the battery’s displacement directions span $\mathbb{R}^9$ with a stated condition number. Nothing in this paragraph is a claim.

Money. Include two stake levels in absolute units of local income and let the monetary coordinate carry the absolute stake. If the recovered $\bar{\Omega}$ has $\bar{\Omega}_{11}$ within noise of zero at the low level and above it at the high level, that is the first direct estimate of what Section 4 declined to claim.

7 Behavioral Game Equilibrium on a Constructed Space

7.1 Definition without circularity

A cost of the form $\|C_i(s) - C_i(s^*)\|^2$ with s^* the candidate equilibrium is zero at every candidate, so every profile is an equilibrium of it. The target must be exogenous to the candidate.

Definition 5 (Behavioral game). A behavioral game has N players with strategy sets $S_i \subset \mathbb{R}^{n_i}$, observers $O_i = (C_i, G_i, B_i)$ with $C_i : S \rightarrow \mathbb{R}^{m_i}$, target maps $t_i : S_{-i} \rightarrow \mathbb{R}^{m_i}$, and, when $B_i < m_i$, projectors Π_i onto a B_i -dimensional subspace of $\mathbb{R}^{m_i}$, with $\Pi_i = I$ otherwise. Player i 's cost is

$$c_i(a_i; s_{-i}) = (C_i(a_i, s_{-i}) - t_i(s_{-i}))^\top \Pi_i^\top G_i \Pi_i (C_i(a_i, s_{-i}) - t_i(s_{-i})). \quad (11)$$

A behavioral game equilibrium (BGE) is a profile s^* with $s_i^* \in \arg \min_{a_i \in S_i} c_i(a_i; s_{-i}^*)$ for every i .

7.2 Existence

Theorem 4 (Existence). *Suppose for every i that S_i is nonempty, compact, and convex, that c_i is continuous on S , and that $a_i \mapsto c_i(a_i; s_{-i})$ is quasiconvex for every s_{-i} . Then a BGE exists. The quasiconvexity hypothesis holds in particular when C_i is affine in a_i and t_i does not depend on a_i , and when C_i is scalar, affine in a_i , and bounded above by a constant target t_i .*

Proof. Let $\beta_i(s_{-i}) = \arg \min_{a_i \in S_i} c_i(a_i; s_{-i})$. It is nonempty because a continuous function attains its minimum on a compact set, convex because sublevel sets of a quasiconvex function are convex and β_i is a sublevel set, and closed-graph by continuity of c_i , hence upper hemicontinuous into the compact S_i . The product correspondence $\beta = \prod_i \beta_i$ on the compact convex S inherits these properties, and Kakutani's theorem [17] gives a fixed point, which is a BGE. This is the Debreu, Glicksberg, and Fan argument [18, 19, 20]. For the first special case, $C_i(a_i, s_{-i}) - t_i(s_{-i})$ is affine in a_i and (11) is a convex quadratic in a_i . For the second, $c_i = (t_i - C_i)^2$ with $C_i \leq t_i$ is a decreasing function of the affine map C_i , so its sublevel sets are superlevel sets of an affine function and are convex. $\square$

7.3 Correspondence with Nash equilibrium

Theorem 5 (Nash correspondence). *Let a behavioral game satisfy the hypotheses of Theorem 4, and define utilities $u_i(a_i, s_{-i}) = K_i(s_{-i}) - c_i(a_i; s_{-i})$ for arbitrary functions K_i of the other players' strategies. Then the BGE of the behavioral game and the Nash equilibria of the game with utilities u_i coincide. Conversely, if a game with utilities u_i that are concave quadratic in a_i with own-strategy Hessian $-H_i(s_{-i})$, $H_i \succ 0$, and own-strategy maximizer $a_i^*(s_{-i})$ is given, then $C_i(a_i, s_{-i}) = a_i$, $G_i = H_i(s_{-i})$, $t_i = a_i^*(s_{-i})$, $\Pi_i = I$ defines a behavioral game whose BGE are the Nash equilibria of that game.*

Proof. For fixed s_{-i} , maximizing u_i in a_i is minimizing c_i because $K_i(s_{-i})$ is constant in a_i , so the best-response correspondences coincide and so do their fixed points. For the converse, a concave quadratic in a_i with Hessian $-H_i$ and maximizer a_i^* equals $u_i(a_i^*, s_{-i}) - \frac{1}{2}(a_i - a_i^*)^\top H_i(a_i - a_i^*)$ exactly, which is of the form $K_i - \frac{1}{2}c_i$ with the stated observer, and a positive factor does not change the argmin. $\square$

Remark. The theorem does not say that every game is a behavioral game. It says that the class of games whose utilities are a function of the others' strategies minus a projected pullback distance to an exogenous target is exactly the class whose Nash equilibria are BGE, and that concave

quadratic games are in it with G_i the negative own-strategy Hessian. For a general concave utility the second-order expansion at the maximizer gives a behavioral game whose BGE approximate the Nash equilibria near the maximizer, and that approximation is local.

7.4 Complexity

Proposition 4 (Hardness). *Computing a BGE is PPAD-hard. For the subclass in which each S_i is a probability simplex, C_i is the multilinear expected payoff of a finite game, $G_i = 1$, $\Pi_i = 1$, and t_i is the constant $M_i = \max_s u_i(s)$, computing a BGE is PPAD-complete.*

Proof. In the subclass, $c_i(a_i; s_{-i}) = (M_i - u_i(a_i, s_{-i}))^2$ with $u_i \leq M_i$, which is quasiconvex in a_i by the second special case of Theorem 4 and is minimized exactly where u_i is maximized. The BGE of the subclass are therefore the mixed Nash equilibria of the finite game, whose computation is PPAD-complete by Daskalakis, Goldberg, and Papadimitriou [21] and by Chen, Deng, and Teng [22]. Hardness of the general problem follows because it contains the subclass. Membership of the general problem in PPAD is not claimed. It requires an approximate-equilibrium formulation with Lipschitz best responses and is left open. $\square$

Proposition 5 (Cost of the structural search that was run). *The structural search of the economics paper evaluated every active set of size at most five over nine dimensions, with a 20-point grid per active variance for sizes one and two and 5,000 seeded draws for sizes three to five. The number of objective evaluations was*

$$\binom{9}{1} 20 + \binom{9}{2} 20^2 + \left(\binom{9}{3} + \binom{9}{4} + \binom{9}{5} \right) 5,000 = 180 + 14,400 + 1,680,000 = 1,694,580,$$

each evaluation scoring nine game targets whose predictions are minima over at most 101 grid alternatives. This is a count, not an asymptotic bound, and it does not include the continuous refinement that a full optimization would add.

Proof. Direct count from the search description in the economics paper's Section V. $\square$

7.5 A closed-form shift of the best response when a direction gains weight

Proposition 6 (Weight on an unread direction moves the best response). *Let a single decision maker choose $x \in [0, 1]$ and let the displacement coordinates be affine, $\Delta a_k(x) = \alpha_k x + \beta_k$ for k in an active set K , with cost $c(x)^2 = \sum_{k \in K} p_k (\alpha_k x + \beta_k)^2$, $p_k > 0$, and interior minimizer*

$$x^* = - \frac{\sum_{k \in K} p_k \alpha_k \beta_k}{\sum_{k \in K} p_k \alpha_k^2}.$$

Add a monetary coordinate with $\Delta a_1(x) = -x$, so $\alpha_1 = -1$ and $\beta_1 = 0$, with weight $p_1 \geq 0$. Then

$$x^*(p_1) = \frac{x^*(0) \sum_{k \in K} p_k \alpha_k^2}{\sum_{k \in K} p_k \alpha_k^2 + p_1},$$

which is strictly decreasing in p_1 whenever $x^(0) > 0$ and tends to 0 as $p_1 \rightarrow \infty$.*

Proof. Differentiate the convex quadratic and solve. The monetary term adds p_1 to the denominator and nothing to the numerator because $\beta_1 = 0$. $\square$

Remark. In the economics paper’s ultimatum encoding the social and identity coordinates are affine in the offer for offers below one half, the monetary displacement is minus the offer times the acceptance probability, and the epistemic coordinate carries the rejection risk. Proposition 6 is the linear core of that encoding with the rejection term frozen. It says that giving any positive weight to the monetary coordinate lowers the offer, and by more as the weight grows, which is the direction of the Andersen data, where mean offers fall from 24 percent at the lowest stake to 12 percent at the highest. It is a best-response statement for the proposer under a fixed responder rule. It is not a fixed-point computation of a two-player BGE, and the economics paper reports no such computation. A two-player version needs the responder’s target and observer written down and its fixed point computed, which is future work.

8 Independent Use

Three things would let outside researchers use the construction without adopting its vocabulary. The first is the readscope instrument [14], already released, which recovers a read operator from consumer calls, certifies its rank, and reports the budget cliff. The second is the eris-econ package [9], already released, which reproduces every number of the economics paper from the encodings of its Table IV. The third is one new domain in which the consumer is a black box and the output metric is a success rate, so that the read operator is recovered rather than coded and the transfer regret of Proposition 1 is measured between two consumers of the same representation. A tool-selection consumer inside a language-model agent, with the embedding of the tool description as the source and task success as the metric, is one such domain, and it is a measurement to be registered, not a claim.

9 What Is Proved Here and What Is Not

Statement	Status	Where
Fibers of the consumer are the indistinguishability classes; regular fibers are submanifolds with tangent $\ker J_C$; quotient is a manifold for a submersion	proved	Theorem 1(a)
$P_C = J_C^\top G J_C$ is PSD with kernel $\ker J_C$; definite iff J_C injective; averaged kernel is the a.e. common kernel	proved	Theorem 1(b)
Global consumer-relative distortion via d_G , with the quadratic form as its second-order expansion	proved	Theorem 1(c)
Transfer regret is nonnegative; closed form with Ky Fan misalignment in the isotropic linear case	proved	Theorem 1(d), Proposition 1
Consumer-relative rate-distortion function is nonincreasing and convex; Gaussian water-filling on $P^{1/2}\Sigma_X P^{1/2}$	proved	Proposition 2
Rate-content region	defined	Definition 3
Local divergence geometry of a channel that reads through C is the read operator with the Fisher metric as G	proved, local	Theorem 2
Conditional mutual information equals an averaged read operator	not claimed	Remark after Theorem 2
The coding paper's region is the consumer-relative region for the linear-functional consumer	identification of definitions, operational constraints verified item by item	Corollary 1 and Lemma 1
The economics paper's cost is a pullback distance with the encoding as consumer	proved	Corollary 2
Unread money implies stake invariance, and the converse under a stake-carrying encoding	proved	Proposition 3
A budget cliff at which money enters	not claimed	Section 4
Budgeted menu-relative observer violates regularity and is outside every RUM	proved	Theorem 3
The fitted economics model is outside RUM	false, and not claimed	Section 5
Learned-geometry protocol with cliff-derived design bound	protocol	Section 6
BGE with exogenous target exists under compactness, continuity, quasiconvexity	proved	Theorem 4
BGE equals Nash for $u_i = K_i - c_i$ and for concave quadratic games	proved	Theorem 5
BGE is PPAD-hard; complete on the finite-game subclass; membership in general open	proved as stated	Proposition 4
Evaluation count of the search that was run	proved	Proposition 5
Weight on the monetary direction lowers the best-response offer in the affine core	proved	Proposition 6
Two-player BGE shift matching Andersen	not computed	Section 7

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